

PRAJAKTA DHANANJAY BHOSALE

Senior Software Engineer | Automotive Electronics | ADAS Systems | Onsite Japan

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PROFESSIONAL SUMMARY

Automotive Senior Software Engineer around 5 years of strong experience in system-level engineering, requirement understanding, and validation in ADAS and EE architecture projects. Skilled in ADAS feature understanding, CAN communication, and IBM Rhapsody tool. Seeking a System Engineer role in Japan where I can contribute technical knowledge of automotive systems while leveraging my Japanese language skills enabling smooth technical and cultural communication.

PROFESSIONAL EXPERIENCE

KPIT Technologies, Pune, Maharashtra, India | Onsite: Japan

Senior Software Engineer (Onsite — Japan)

Fusion Data Analysis Project

October 2025 – Present

- To analyze and validate fused object output from multiple sensors (camera & radar) to ensure correct detection, classification, tracking, and confidence performance for ADAS functions.
- Performed fusion output analysis for detected objects (vehicles, pedestrians, lane-related objects).
- Validated object attributes such as position, speed, distance, ID tracking, confidence, and classification.
- Identified mismatches between sensor outputs and fusion results and reported anomalies.
- Coordinated with Japan stakeholders and offshore teams for smooth project execution.

TSR (Traffic Sign Recognition) Data Analysis Project

June 2024 – September 2025

- Conducted TSR feature verification and validation using real-world driving datasets. Performed sign detection analysis using Google Maps integration for ground truth comparison.
- Generated technical reports highlighting detection accuracy, false positives, and missed signs along with possible error detection.
- Automated analysis tasks using Python to improve reporting efficiency and reduce manual effort.
- Prepared detailed analysis reports for stakeholders to support decision-making for TSR feature for the due diligence activity.

Software Engineer (offshore — Pune)

Camera Data Analysis Project (Honda Account)

January 2024 – May 2024

- Camera – Data Tagging & Algorithm Support- Performed scene tagging and camera dataset analysis using FSViewer.
- Supported camera performance improvement through systematic data classification. Worked on dirt detection software improvement using Qride software.
- Delivered analysis results and documentation for internal and client review.

- Bridged technical and language gaps, enabling efficient cross-cultural execution.

Software Engineer (offshore — Pune)

ECAF (Electrical and Electronic Architecture Framework) Project (PACCAR Account) *March 2021 – December 2023*

- Gained 2 years 10 months of hands-on experience on truck-level E/E architecture.
- Supported automotive ECU level communication protocols using CAN and J1939.
- Gained knowledge of signal-level requirement documentation understanding - TRS,FRS,FDS.
- Performed CFA signal mapping activities for truck ECUs and network communication. Contributed to FBD (Function Block Diagram) generation and system flow representation.
- Created architecture diagrams and mappings using IBM Rational Rhapsody, MS Visio, and YED GraphML.
- Managed requirements and technical documentation using Windchill, IBM DOORS Next (DNG) and version control tool Tortoise SVN.

TECHNICAL SKILLS

Programming Languages & Development Tools:

C Programming, Python, Basics of MATLAB & Simulink, Data Analysis, Automation Scripting

Automotive Technologies & Protocols:

CAN and J1939 Protocol, ADAS features, ADAS sensors, V-Model lifecycle

Software Tools & Platforms:

IBM Rational Rhapsody, FSViewer, Microsoft Visio, SVN Tortoise, Windchill, DNG, YED GraphML

Professional & Soft Skills:

Bilingual Communication (Japanese–English), Stakeholder Requirement understanding, Technical Documentation, Team co-ordination/management

LANGUAGES

- **English:** Professional Working Proficiency
- **Marathi:** Native Language
- **Japanese:** JLPT N3 Certified, Business Japanese N2 (In Progress)

EDUCATION

Bachelor of Engineering (Electronics)

KIT College of Engineering, Kolhapur

87.13% | 2017–2020

Engineering Diploma (Electronics & Telecommunication)

Government Polytechnic Kolhapur

86.69% | 2014–2017

Secondary School Certificate (SSC)

Princess Padmaraje Girls High School, Kolhapur

86.80% | 2014

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- Generative AI and Prompt Engineering — KPIT Academy (2025)
- Advance Driver Assistance Systems — Udemy (2024)
- ADAS Sensors - Udemy (2024)
- Python Programming Basics — Udemy (2024)
- MATLAB and Simulink Onramp Certification — MathWorks Platform (2023)
- ROS (Robotics Operating System) Workshop — MIT University Pune & RSPinJapan (2023)
- ECODE Automotive Courses — CANoe and CAPL scripting & Embedded Electronics (2022)
- PACE and Genesis Training — KPIT Academy (2021)

AWARDS & RECOGNITION

- **Client Excellence Award (2024)** — Honda Account for outstanding project performance
- **Client Excellence Award (2023)** — PACCAR Account for exceptional contributions
- **Krown Award "The Comrades" (2023)** — Team recognition for collaborative excellence in PACCAR projects
- **KPCL Cricket Champions (2023)** — Women's champions trophy winner in KPIT PACCAR Cricket League
- **CSR Volunteer** — Active participant in "Chote Scientist" community engagement event at KPIT

HOBBIES & INTERESTS

- **Warli Painting** — Traditional Indian folk art form
- **Trekking** — Adventure and nature exploration
- **Cross-cultural Exchange** — Learning about different cultures and languages